Chuhanwen (Silas) Sun

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Education

Rutgers University New Brunswick, NJ, USA
Ph.D. Student, Graduate Program in Microbiology and Molecular Genetics *Sept. 2025 – Present*

Karolinska Institutet Stockholm, Sweden
Master of Biomedicine, Pass with distinction *Aug. 2023 – June 2025*

Tsinghua University Beijing, China
Bachelor of Life Sciences, GPA: 3.67 *Sept. 2019 – Aug. 2023*

Awards and Scholarships

Bridge to Excellence Award Sept. 2025
Rutgers University, USA — $40,000 doctoral fellowship

Excellent Scholarship of Comprehensive Quality Sept. 2023
Tsinghua University, China — $1,000 for academic excellence

Second-class Prize Scholarship Sept. 2022
Tsinghua University, China — $1,000 for contributions in student organizations

Publications

Peer-Reviewed Publications

- **Sun, C.**, Zhang, Y. (2025). STHD: probabilistic cell typing of single spots in whole transcriptome spatial data with high definition. *Genome Biology*, 26, 213. DOI: 10.1186/s13059-025-03608-4
- Zhuang, Y., Li, Z., Xiong, S., Sun, C., Li, B., Wu, S.A., **Sun, C.**, et al. (2023). Circadian clocks are modulated by compartmentalized oscillating translation. *Cell*, 186(15), 3245–3260.e23. DOI: 10.1016/j.cell.2023.05.045

Preprints / Under Review

- **Sun, C.**, Thomas, R., Stringer, C., et al. (2026). Epigenetically constrained astrocyte states underlie prefrontal cortex vulnerability in Down syndrome-associated Alzheimer's disease. *bioRxiv preprint.* DOI: 10.64898/2026.04.17.719050
- **Sun, C.**, Eroglu, A., Hausser, J. (2025). bioLUCID: quantifying batch and biological effect in single-cell transcriptomics by hypothesis-driven variance decomposition. *Nature Communications*, under review.
- Chen Wu, L., Hu, X., Zhan, F., **Sun, C.**, Gonzales, J., Ofer, R., Tran, T., Verzi, M.P., Liu, L., Yang, J. (2026). STCS: A platform-agnostic framework for cell-level reconstruction in sequencing-based spatial transcriptomics. *bioRxiv preprint.* DOI: 10.64898/2026.02.26.708370
- Wu, Y., Zhang, T., Zhou, H., Wu, H., **Sun, C.**, et al. (2024). DeepCRE: AI-driven cross-drug response evaluation to transform drug R&D. *arXiv preprint arXiv:2403.03768.* Under review at *Science Advances.* DOI: 10.48550/arXiv.2403.03768
- Jia, J., Liu, M., Xi, X., **Sun, C.**, et al. (2024). Spatiotemporal dynamics of mononuclear phagocyte cells reveals new regulators of plaque instability in atherosclerosis. Under review.

Invited Talks and Presentations

Accelerating Research for Down Syndrome Apr. 9, 2026
Invited Speaker, Alana Down Syndrome Center – University of São Paulo Webinar *Online*

- Presented research as a Rutgers University Yang Lab graduate student in a multi-institutional webinar featuring speakers from MIT, University of São Paulo, and Rutgers University.

Research Experience

Department of Cell and Molecular Biology, Karolinska Institutet Stockholm, Sweden
Graduate Research Assistant *Oct. 2023 – Sept. 2025*

- **Quantitative Assessment of Batch Effects in scRNA-seq Data** *supervised by* Prof. Jean Hausser
  - Developed a hypothesis-driven variance decomposition framework, **bioLUCID**, to distinguish technical batch effects from biological signal in single-cell transcriptomic integration.

- Proposed a quantitative metric, the **qsh/qsp ratio**, to independently measure batch-driven and biological variation on a common benchmarking scale.
- Designed cross-dataset and simulation benchmarks showing that bioLUCID detects over-correction effects missed by existing integration diagnostics.

- **Spatial Modelling in Triple-Negative Breast Cancer (TNBC) Data** *supervised by* *Prof. Jean Hausser*
  - Built an interpretable spatial modeling framework, the **Tumor Promoting Microenvironment Characterization (TUPMIC) model**, to quantify tumor-promoting microenvironmental features.
  - Engineered patient-level features from **multiplexed imaging data** across 41 TNBC patients and modeled their associations with tumor growth using β-regression.
  - Interpreted model outputs to recapitulate macrophage and monocyte associations and prioritize candidate T cell phenotypes with potential anti-tumor effects.

Department of Neurosurgery, Duke University — Durham, NC, USA

Short-term Visiting Scholar — *Apr. 2024 – Sept. 2024*

- **Probabilistic Cell Typing in Whole Transcriptome Spatial Data** *supervised by* *Prof. Yi Zhang*
  - Co-developed a probabilistic **machine learning** method, **STHD**, for cell typing in high-resolution whole-transcriptome spatial data, leading to publication in *Genome Biology* (2025).
  - Designed benchmarking analyses comparing STHD with existing spatial cell-typing methods across simulated, public, and patient-derived VisiumHD datasets.
  - Implemented a spatial cell-type mapping and biological interpretation workflow for VisiumHD data using high-definition (2μm) cell-type assignments.

Department of Automation, Tsinghua University — Beijing, China

Summer Research Assistant — *Jun. 2023 – Jul. 2024*

- **Spatiotemporal Analysis of Unstable Atherosclerotic Plaques** *supervised by* *Prof. Xuegong Zhang*
  - Integrated **scRNA-seq and spatial transcriptomics data** to reconstruct plaque microenvironment dynamics and identify cell states associated with instability.
  - Identified a disease-associated mononuclear phagocyte cell (MPC) state and prioritized EGR1 as a candidate biomarker and therapeutic target for plaque instability.
- **Consistency Analysis in Vascular Remodeling Diseases (VRDs)** *supervised by* *Prof. Xuegong Zhang*
  - Built a reproducible **scRNA-seq analysis pipeline** for cross-disease integration of 17 samples across 7 vascular remodeling diseases.
  - Combined **SCENIC** and **protein-protein interaction (PPI)** analyses to compare MPC regulatory and differentiation programs, revealing shared and disease-specific patterns.

AirTsinghua — Beijing, China

Research Internship — *Jul. 2022 – Aug. 2023*

- **Data Preparation for a Deep Learning Model Predicting Drug Response** *supervised by* *Prof. Zaiqing Nie*
  - Curated and harmonized public **multi-omics data (RNA-seq, methylation, CNV)** to construct a patient-level training dataset for cross-drug response evaluation (CRE).
  - Integrated **cancer cell line sensitivity data** from six public databases and used **Markov Chain Monte Carlo (MCMC)** curve fitting to normalize drug-response measurements across platforms.
  - Generated analytical outputs and contributed figures and methods text to the arXiv preprint.

IDG/McGovern Institute, Tsinghua University — Beijing, China

Undergraduate Research Assistant — *Mar. 2022 – Sept. 2023*

- **Decode Pathophysiological Roles of ATXN2/2L in Neuronal Systems** *supervised by* *Prof. Yi Lin*
  - Generated and validated cellular perturbation models for ATXN2/2L biology, including 16 stable U2OS double/single knockout cell lines with polyglutamine (polyQ) length variation.
  - Purified ATXN2/2L proteins with varying polyQ lengths and used **in vitro phase separation** assays to test how polyQ expansion alters condensate behavior.
  - Developed a cerebellum-specific ATXN2/2L knockout mouse model through **Cre-loxP** and performed behavioral phenotyping of motor function and balance.
  - Analyzed cell-staining confocal images and contributed visualizations supporting publication in *Cell*.

- **Effect of Clock Protein Phase Isolation on Rhythmicity** *supervised by* *Prof. Yi Lin*
  - Engineered Clock protein mutants to test how protein truncation affects phase separation behavior and

- circadian rhythmicity.
- Generated and validated a Clock-Flag knock-in U2OS cell line using CRISPR/Cas9, Western Blot, immunostaining, and DNA sequencing.

Tsinghua Laboratory of Brain and Intelligence Beijing, China
Undergraduate Research Assistant *Jul. 2021 – Dec. 2021*

- **Modeling and Behavioral Testing of a Parkinson's Disease Mouse Model** *supervised by Prof. Yichang Jia*
  - Established and validated a neurodegeneration **mouse model** for Parkinson's Disease using MPTP injection and rotenone administration.
  - Evaluated disease phenotypes through behavioral testing (Open Field and Rotarod) and immunofluorescent staining of substantia nigra tissue.
- **Study of Learning and Memory Behaviors in CX3CR1 Knockout Mice** *supervised by Prof. Yichang Jia*
  - Engineered a CX3CR1 knockout mouse model via **Cre-loxP** and designed behavioral assays to assess learning and memory.
  - Prepared whole-brain dissection and **RiboTag bulk RNA sequencing** samples to investigate translatomic changes during learning and memory.
  - Performed quantitative image analysis of hippocampal microglia morphology and abundance using **ImageJ**.

Skills

Computational Biology and Omics:
- Single-cell RNA-seq analysis; spatial transcriptomics analysis (Xenium, Visium); multiplexed imaging analysis; multi-omics data integration.

Statistical and Machine Learning Methods:
- Variance decomposition and batch-effect assessment; probabilistic modeling and cell-type inference; regression modeling; MCMC-based fitting; benchmarking, simulation, and reproducible pipeline development.

Software Packages and Tools:
- Python, R; NumPy, pandas, PyTorch; Scanpy, Squidpy, Seurat, ArchR.

Experimental and Model Systems:
- Mouse models and behavioral testing; stable cell line construction; gene editing (CRISPR/Cas9, Cre-loxP); molecular techniques (Immunohistochemistry, Western Blot, PCR).

Languages:
- English (IELTS: 7.5).